Joint RFID and UWB Technologies in Intelligent Warehousing Management System

Kang Zhao, Minghua Zhu, Bo Xiao, Xuguang Yang, Changlei Gong, and Junyi Wu

***Abstract*—Joint application of radio-frequency identification (RFID) and ultrawideband (UWB) technologies in the intelligent warehousing management system is proposed. In this system, we regard forklift as infrastructure, and both the UWB mobile terminal (MT) and the RFID reader are mounted on the forklift. The RFID reader is used not only to read the information of goods but also to determine the goods' status of loading and unloading. The UWB MT is used to locate the forklift. The goods or pallets are labeled with RFID tags. Utilizing the integration of these two technologies, the dual goals of goods information and goods location perception are achieved. An *M*/*N*-*K* sliding window method is proposed to determine the loading and unloading of goods in this article. Our experiments reveal that this novel method can quickly, accurately, and efficiently determine the states of the goods on the forklift. For the indoor localization, an algorithm is proposed based on RSS residual weighting (RRW). Experiments show that RRW can mitigate the nonline-of-sight error substantially compared with the conventional Taylor algorithm and recent Convex approximation algorithm. Finally, a real working practice in a warehouse of a company is introduced, and it illustrates the feasibility of the system.**

***Index Terms*—Indoor positioning, intelligent warehousing management system (IWMS), Internet of Things (IoT), NonLine of Sight (NLOS), radio-frequency identification (RFID), received signal strength (RSS), ultrawideband (UWB).**

I. Introduction

IN THE early 1990s, with the increase of global manufacturing competition, people began to point out the automation of warehouse design to achieve high throughput and productivity [1], and to reduce order processing costs [2], [3]. There has been an international trend to focus on the development of warehousing management since the 21st century [4]. Researches show that the order picking activity is one of the most time-consuming processes of a warehouse [5], [6]. The order picking activity is estimated to contribute to more than 55% of the total cost of warehouse operations [7]. For this reason, experts proposed various technologies to solve this problem [8], [9], among which the radio-frequency identification (RFID) [10], [11] has become one of the indispensable technologies.

In the past decade, many experts and scholars studied the application of RFID in intelligent warehousing management. Chow *et al.* [12] proposed an RFID-based resource management system for warehouse operations with a time saving and cost-effective approach. The proposed system integrated the case-based reasoning techniques and RFID technique. Jungk *et al.* [13] described how RFID was used to track and trace the transportation of goods by forklift. They mainly discussed the system structure. Additional sensor (photo sensor) is required in their scheme to determine whether there were goods on forklift. Bo *et al.* [14] put forward a warehousing management system based on RFID, and mainly probed into the structure and operation flow of the system. Wang *et al.* [15] proposed a digital warehousing management system (DWMS). By using the RFID technology, the DWMS enables a plane warehouse to achieve visualized inventory management, automatic storage/retrieval assignment, and high accuracy of inventory control. They had applied their system to a tobacco company in China. Qian *et al.* [16] designed an RFID-based automated warehousing system. They constructed a whole new storage module, which includes storage array, storage/output conveyors, tote pusher with the RFID reader, and so on. It had the features of scalability, reconfigurability, and convenience of goods positioning. However, their scheme needed to overthrow the traditional design of the warehouse, and the cost is high. Modrak *et al.* [17] tried to install the RFID reader on the forklift and the entrance. They discussed the information reading of goods and the judgment of entry and exit. Li *et al.* [18] installed antenna at the bottom of the forklift besides in front of the forklift. In their scheme, RFID tags were also installed on the floor of the warehouse, and the bottom antenna of the forklift was used to sense the tags on the floor for positioning purposes. The system had been successfully implemented in the distribution center of Bailian Group, which is one of the biggest retail companies in China. Lee *et al.* [19] developed a data collecting unit (DCU) integrated with various sensors, such as RFID and ultrawideband (UWB). The DCU was installed on forklift, and the system can be applied to intelligent warehousing management, while they said nothing more about data processing of RFID and UWB.

Because the radio signal of RFID is susceptible to interference, its original data are not always reliable. The errors occur in the process of data capturing, including false

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negatives (FNs), false positives (FPs), and duplicates [20]. FN is that tags are mistakenly assumed to have exited the reader's detection range. FP is that tags are mistakenly assumed to be presented after they have exited the reader's detection range. *Duplicate readings* occur when tags are in the range of a reader and are read multiple times or when they are read by multiple readers. Of all the errors, FNs are the most common and serious. According to statistics, the general loss rate of RFID data can reach about 30% [21], which is the primary factor limiting the widespread adoption of the RFID technology [22]. Consequently, it is necessary to consider the cleaning of RFID data. Many algorithms were developed in this context. Jeffery *et al.* [23] proposed an extensible stream processing (EPS) scheme, which was more suitable for multiple sensors. The slide window was proposed for the first time in 2006 [24]. But the static size of the window was the limitation of this approach, because large window induced FPs and a small window cannot fill FNs. To solve this problem, a powerful algorithm statistical smoothing for unreliable RFID data (SMURF) was proposed in [21]. SMURF exploited a view of RFID unreliability by modeling observed RFID readings as an unequal-probability random sample of tags in the physical world. It was designed to be a component in a pipeline of operators responsible for low-level RFID data processing tasks, such as cleaning, filtering, and spatial processing. SMURF has weaknesses in dealing with the size of slide window. When the values of window size computed by equations in SMURF are in conflict, it is difficult to choose the window size [22]. Another weakness is that SMURF cannot adapt to fast-moving tags. Meng and Yu [22] proposed an improved algorithm based on SMURF, considering the speed of tag. Many bloom-filter-based approaches for filtering duplicate RFID data streams were proposed in [25]–[28]. But these approaches for filtering duplicate data were complex and slow as they use multiple hash functions. A Bayesian inference approach was used in [29] to cope with real-time RFID data streams. They had developed an n-state model for 2-D RFID reader arrays and computes the likelihood of tags presence exploiting environmental constraints to improve data cleaning. This method gave good results but it requires many readers and a good knowledge of the environment. Baba *et al.* [30] proposed a regular expression matching-based approach to find all the matching paths that trigger matching from raw RFID data. It actually referred to the path estimation method using RFID in a specific application environment. Mihoub-Sidi-Moussa *et al.* [31] disscussed more about the estimation of the reading rate by RSSI value, and not much about the cleaning method.

In terms of indoor positioning, many technologies have been explored, including acoustic, magnetic, optical, laser, and radio-frequency-based technologies [32], [33]. Digital microelectro-mechanical system (MEMS) audio sensors were used in [34], but it only gave a small range (smaller than 5 m $\times$ 5 m) of positioning results. The magnetic sensor was used in a recent paper [35], and it claimed to be able to achieve positioning accuracy less than 4 m. Ali and Mailah [36] outlined some of the recent efforts made toward laser sensors for displacement, distance, and position. There are also many authors using RFID for positioning [37]–[42]. Motroni *et al.* [43] presented a novel phase-based localization method for mobile nodes equipped with an ultrahigh frequency (UHF) RFID system. However, under this method, reference tags must be deployed every 2 m^2. In practice, the positioning accuracy of the RFID solution is not ideal. UWB-based communication and positioning systems have become popular as UWB signals possess the capability of measuring accurate position up to centimeter level due to its high multipath resolution [44]–[46]. However, there are some challenges to implement these systems in an indoor area with obstacles and barriers. One of the challenges is NonLine-of-Sight (NLOS) conditions between two communicating nodes which results in large localization error [47]. This is inevitable in the warehouse environment. It is expected to be a serious barrier to UWB indoor positioning, and it is a great challenge to solve NLOS directly by the mathematical method. Güvenç *et al.* [48] proposed an NLOS identification technique based on the multipath channel statistics, such as the kurtosis, the mean excess delay spread, and the root mean-square delay spread. Maranò *et al.* [49] developed classification and regression algorithms based on support vector machine (SVM), which were capable of distinguishing LOS/NLOS propagation and further mitigating the ranging error in NLOS conditions. However, these two methods [48], [49] relied on the analysis of the received waveform, which was not practical. Chen [50] introduced a residual weighting algorithm (Rwgh) providing a way of reducing the NLOS errors conveyed into the location estimate without prior information on error models. Xu *et al.* [51] used the residual to select the primary AN (UWB Anchor) for time difference-of-arrival (TDOA) location, which did not require prior knowledge and can mitigate NLOS error. Silva and Hancke [52] investigated impulse radio ultrawideband (IR-UWB-based) ranging and NLOS identification in industrial environments, but their approach was not applicable to all environments. Wang *et al.* [53] developed a novel robust Convex approximation method for TDOA-based localization under NLOS conditions. The key feature of the approach is that it does not require knowledge of the distribution or statistics of the NLOS errors.

Aiming at the application of intelligent warehousing management and considering the tradeoff between cost and accuracy, we adopt a novel scheme to fulfill the goods sensing and location awareness. On the one hand, we install the RFID reader on the forklift, and attach the RFID tags (about 0.1 RMB each) on the packaging of goods or pallets. The RFID reader not only reads the goods information but also determines the loading and unloading status of the goods. The existing methods cannot satisfy the RFID cleaning requirements in the new situation. We propose a novel loading and unloading decision algorithm for RFID data stream cleaning, which is proved to be simple, efficient, and has high real-time performance. On the other hand, we make use of the UWB technology to locate the forklift to indirectly know the location of the goods, and avoid the installation of UWB tags (about 200–400 RMB each) on each goods. We integrate the goods decision module and the UWB tag into an intelligent mobile terminal (MT). Consequently,

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both the RFID reader and intelligent MT are installed on the forklift.

A warehouse environment is a typical harsh environment. In order to mitigate the ranging error in NLOS conditions, the RSS residual weighting algorithm (RRW) is proposed in this article. By the use of the ranging measurement (such as TDOA) the position is estimated, then a theoretical received signal strength (RSS) value can be estimated, and we exploit the residual between the estimated RSS and the measured RSS to indicate the degree of NLOS. Assuming that the number of ANs is greater than the minimum number required during ranging measurements, we can use RSSI residual as the weight of each location result, which is obtained from different ANs combinations.

More concretely, the contributions of this article can be summarized as follows.

1) A novel intelligent warehousing management scheme combining RFID and UWB technologies is proposed, and it makes a tradeoff between cost and accuracy of positioning.
2) We propose a novel RFID data cleaning algorithm using M/N-K sliding window, which can accurately judge the loading and unloading status of goods.
3) A novel algorithm named RRW is proposed, and it can mitigate the NLOS error to enhance the accuracy of positioning.

The remainder of this article is organized as follows. In Section II, the structure of the system is presented. In Section III, two core algorithms are introduced: 1) the loading and 2) unloading identification algorithm of the goods based on M/N-K sliding window and the RRW algorithm for location awareness. In Section IV, the experimental results of the goods sensing algorithm and the simulation results of the RRW algorithm in NLOS condition are given. Subsequently, a real case is introduced. Finally, this article is concluded and some future works are discussed in Section V. The list of acronyms and abbreviations used in this article are summarized in Table I.

TABLE I
ACRONYMS/ABBREVIATIONS

AN	UWB Anchor
BRI	Basic Reader Interface
CDF	Cumulative Distribution Function
DWMS	digital warehousing management system
EPC	Electronic Product Code
ERP	enterprise resource planning system
FN	False Negative
FP	False Positive
GPS	Global Positioning System
ISO	International Organization for Standardization
IWMS	Intelligent Warehousing Management System
LOS	Line-of-Sight
MT	UWB mobile terminal
NLOS	Non-Line-of-Sight
RFID	Radio Frequency Identification
RMSE	Root Mean Square Error
RRW	RSS residual weighting algorithm
RSS	Received Signal Strength
SMURF	Statistical Smoothing for Unreliable RFID data
TDOA	Time Difference Of Arrival
TN	True Negative
TP	True Positive
UHF	Ultra High Frequency
UWB	Ultra Wideband

II. System Design and Architecture

A typical RFID installation consists of three components: 1) readers; 2) antennas; and 3) tags. As we hope that RFID has a long detection range, we focus on the 915-MHz RFID technology. As we know that outdoor positioning technologies, such as global positioning system (GPS) cannot perform well due to signal blocking in building. In comparison with the outdoor environments, sensing location information in indoor environments requires a higher precision and it is a more challenging task in part because various objects reflect and disperse signals. UWB is an emerging technology in the field of indoor positioning that has shown better performance compared to others. Consequently, we consider the UWB positioning technology here. Thus, the system developed by us are given as follows.

1) *UHF RFID Tags:* They are attached on the packaging of goods or pallets. Each tag contains a unique identification number that represents specific goods. As RFID tags can be easily damaged during transportation, usually they need to be resistant to impact and anti-extrusion.
2) *UHF RFID Readers and Antennas:* They are mounted on the forklift to read RFID tags.
3) *UWB Anchors:* As reference base stations for positioning, they are usually installed on the top of the warehouse or on the wall. They also include an Ethernet module to transfer TDOA data to a server. In this article, we use the abbreviation AN for UWB anchor.
4) *Intelligent MTs:* They also mounted on the forklift. The terminal includes two functions: a) UWB signal transmission and b) goods sensing (reading RFID tags and the decision of loading and unloading). Note that we also use UWB to transfer the goods data. Since the intelligent terminal is an MT, we also use the abbreviation MT for it.
5) *Background Management System:* The background management system mainly includes positioning engine, goods information processing program, enterprise resource planning system (ERP), and so on. Note that we do not pay much attention to the background management system in this article.

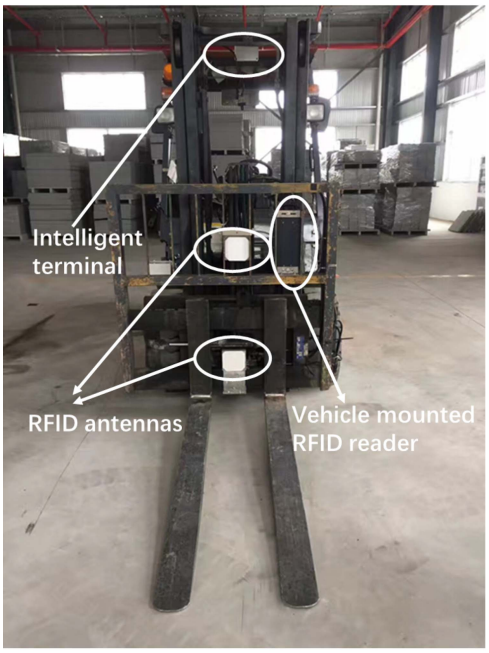
From this composition, we can see that the forklift exists as infrastructure in the whole system. Fig. 1 gives the components on forklift. Equipments installed on the forklift include: 1) RFID reader; 2) RFID antennas; and 3) MT.

For RFID reader, IV7 [Fig. 2(a)] is used. It is produced by *Intermec*© (it has been acquired by *Honeywell*), and it is a vehicle-mounted device. This product can read several types of RFID tags, including ISO 18000-6b, EPC UHF Gen 2, and

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(a)



(b)

Fig. 1. Components on forklift. (a) Illustration of components on forklift. (b) Components on forklift.



(a) (b)

Fig. 2. RFID reader and antenna. (a) Intermec IV7. (b) Laird S9025PL.

EPC Class. Moreover, it can connect up to four antennas. As for antenna, Laird S9025PL [Fig. 2(b)] is used in this article. See Table II for specific parameters of Laird S9025PL. In addition, ANs (Fig. 3) and MT are self-developed equipments. In hardware, the MT is basically the same as the AN, only adding a serial interface with IV7, which is used to gather RFID data from the RFID reader.



Fig. 3. UWB Anchor.

TABLE II
LAIRD 9025 PARAMETERS

Frequency Range	902-928MHz
Gain	5.5dBi
Beamwidth	E-Plane 100°, H-Plane 100°
SWR	≤ 1.5
Polarization	Circular Polarization
Max Power	10W
Input Impedance	50Ω

In the remainder of this section, we will briefly explain the operation process of the intelligent system. When the forklift is electrified, IV7 will constantly read whether there are RFID tags (goods) on the forklift. Simultaneously, the MT will also continuously send positioning measurements to the positioning engine. Once the loading and unloading events occurred, the MT will immediately send the events to the server, i.e., the goods information processing program. The positioning engine will continuously parse the data from MT and calculate the position of it, i.e., forklift. In this way, the background system (e.g., the ERP system) can clearly know the loading position and unloading position of the goods, which provides a fine support for the location management of the goods. Actually, once the goods are loaded onto the forklift, the trajectory of the goods can be known. What is more, the forklift drivers do not have to leave their forklift to scan RFID tags manually or do anything else. They just need to concentrate on driving. Technically, central to the study of this subject are the dynamic sensing of the goods and the location awareness of the goods.

III. CORE SENSING ALGORITHMS

As mentioned in the previous section, goods sensing and location awareness are the core issues of the whole system. For each of these two issues, we will propose two novel algorithms in this section.

A. Goods Sensing

The deployment of the RFID sytem will generate mass of RFID data, and those data streams are often with multiple errors, including the dropped readings, wrong readings, data redundancy, etc. Due to the noncontact feature of the RFID

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technology, radio-frequency signals susceptible to environment interference and inherent conflict between signals, the dropped readings becomes the most serious type error [54]. In this article, the M/N-K sliding window decision method is proposed to clean the RFID data. In the early days, it was used in the detection of object existence in radar and sonar applications [55]. According to the principle of the M/N-K sliding window decision, if M or more successful results occured within N *interrogation cycles*, the target is considered to "exist," i.e., goods have been loaded on the forklift. We call the window in this process M/N window. Similarly, if there is no result for K consecutive times within K *interrogation cycles*, the target is considered to have "disappeared," i.e., goods have been unloaded. We call the window in this process K window. An *interrogation cycle*, which is regarded as a window unit, is one iteration through the reader's protocol that attempts to determine all tags in the reader's vicinity [21]. The length of one interrogation cycle of IV7 is 25 ms, which was obtained through software log analysis.

Briefly, the rule of loading and unloading decision of goods is as follows.

1) The RFID reader continuously reads RFID tags.
2) If the previous status of a certain tag is unloaded and if the number of times this tag is successfully read exceeds or equals to M during the recent N consecutive readings (i.e., the window size is N), the goods represented by this RFID tag are considered to be on the forklift. Once the intelligent terminal detects this condition, it will emit a loading event.
3) If the previous status of a certain tag is loaded and if this RFID tag is not read for K consecutive times, the goods represented by this RFID tag are considered to be unloaded. An unloading event will occur when this change is detected by the MT.
4) In other cases, the status of the goods remains unchanged, that is, the previous state is maintained.

Let us discuss the M/N window rule first, and the K window rule is similar. Generally, we like to use a binary sequence to record the result of each reading [55], [56], and MT maintains a binary sequence for each RFID tag. The length of the binary sequence is N, which is equal to the size of M/N sliding window. Each bit of the binary sequence indicates whether the tag is read in the corresponding interrogation cycle. "1" indicates that the tag is read, while "0" indicates that the tag is not read. The reading result will be inserted into the lowest bit (i.e., the rightmost of the binary sequence). After each insertion, the MT will execute loading and unloading rules, and after they are finished, this binary sequence will move one bit to the left. The number of "1" contained in a recent window is known as the weight of window. We denote it by the symbol $\#(h)_N$ [57]. Here, h denotes the integer value representing a binary sequence within the window. Therefore, according to the loading and unloading rules, Fig. 4 can be used to depict the state transition diagram of goods.

Then the whole algorithm can be described as Algorithm 1, where the symbol $S(\text{tag}_i)$ denotes the status of the tag i. Notations *LOAD* and *UNLOAD* indicate the loaded and unloaded states, respectively.

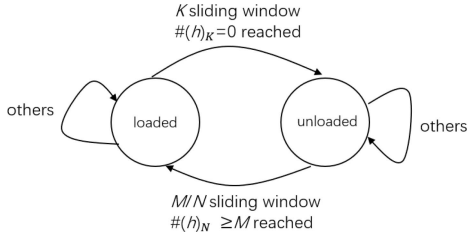


Fig. 4. Goods sensing state transition diagram.

Algorithm 1 M/N-K Algorithm

```
if #(h)_N ≥ M and S(tag_i) is UNLOAD then
    S(tag_i) ← LOAD
    emit a loading event
end if
if #(h)_K = 0 and S(tag_i) is LOAD then
    S(tag_i) ← UNLOAD
    emit an unloading event
end if
```

A tag that was previously not on the forklift is considered to be loaded by the forklift if the condition $\#(h)_N \geq M$ is met. For the RFID tag already on the forklift, an unloading event is triggered if the rule $\#(h)_K = 0$ is met. Obviously, the calculation time of the algorithm has nothing to do with the number of window movements. As long as the last window weights $\#(h)_N$ and $\#(h)_K$ are kept, the calculation of the window weights in the next round is only to add 1 or 0. The rest of the algorithm is simpler, just constant comparisons or assignments. Thus, the entire calculation complexity of Algorithm 1 is $\mathcal{O}(1)$.

Later, we will discuss the determination of M, N, and K values. In this article, we view each interrogation cycle as an independent Bernoulli trial with success probability p. Here, p denotes the reading rate for tag during an interrogation cycle. This implies that the number of successful observations of tag in the window, i.e., $\#(h)_w$, is a random variable that follows a binomial distribution with parameters (w, p), i.e., $\#(h)_w \sim B(w, p)$. Thus, according to the standard probability theory, we can express the expectation and variance of $\#(h)_w$ as

$$E(\#(h)_w) = wp \tag{1}$$

$$\text{Var}(\#(h)_w) = wp(1-p). \tag{2}$$

According to SMURF [21], ensuring that tag is read within the window with probability $> 1 - \delta$ (δ represents the probability that the tag is not observed within the window), the window size w should meet the following condition:

$$w \geq \left\lceil \frac{ln(1/\delta)}{p} \right\rceil. \tag{3}$$

On the other hand, from the aspect of the central limit theorem, the value of $\#(h)_w$ is within $\pm 2Var(\#(h)_w)$ of its expectation with probability close to 0.98 [21]. That is, $\#(h)_w$ has a very high probability (i.e., 0.98) of holding the following condition:

$$|\#(h)_w - wp| \leq 2\sqrt{wp(1-p)}. \tag{4}$$

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Expand (4)

$$wp + 2\sqrt{wp(1-p)} \geq \#(h)_w \geq wp - 2\sqrt{wp(1-p)}. \quad (5)$$

Consequently, it is reasonable for us to set M and N as follows:

$$N = \left\lceil \frac{\ln(1/\delta)}{p} \right\rceil \quad (6)$$

$$M = \left\lceil Np - 2\sqrt{Np(1-p)} \right\rceil. \quad (7)$$

Table III gives the M/N combination under different p and δ values.

Given δ and p, if the window size N set to be $\lceil[\ln(1/\delta)]/p\rceil$, it means the probability of observing the RFID tag within the window is $> 1-\delta$, i.e., the probability of not reading any tag within the window is $< \delta$. Generally, δ is set a very small value. This also means that if the tag cannot be read in the window, then we can think that the tag is outside the reader's vicinity. So K can be set the same as N

$$K = N. \quad (8)$$

TABLE III
VALUE M/N

p \ δ	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.10
0.5	2/10	2/8	2/8	1/7	1/6	1/6	1/6	1/6	1/5	1/5
0.51	2/10	2/8	1/7	1/7	1/6	1/6	1/6	1/5	1/5	1/5
0.52	2/9	2/8	1/7	1/7	1/6	1/6	1/6	1/5	1/5	1/5
0.53	2/9	2/8	2/7	2/7	1/6	1/6	1/6	1/5	1/5	1/5
0.54	2/9	2/8	2/7	1/6	1/6	1/6	1/5	1/5	1/5	1/5
0.55	2/9	2/8	2/7	1/6	1/6	1/6	1/5	1/5	1/5	1/5
0.56	3/9	2/7	2/7	1/6	1/6	1/6	1/5	1/5	1/5	1/5
0.57	3/9	2/7	2/7	1/6	1/6	1/5	1/5	1/5	1/5	1/5
0.58	2/8	2/7	2/7	2/6	2/6	1/5	1/5	1/5	1/5	1/4
0.59	2/8	2/7	2/6	2/6	2/6	1/5	1/5	1/5	1/5	1/4
0.60	3/8	2/7	2/6	2/6	1/5	1/5	1/5	1/5	1/5	1/4
0.61	3/8	2/7	2/6	2/6	1/5	1/5	1/5	1/5	1/4	1/4
0.62	3/8	2/7	2/6	2/6	1/5	1/5	1/5	1/5	1/4	1/4
0.63	3/8	2/7	2/6	2/6	1/5	1/5	1/5	1/5	1/4	1/4
0.64	3/8	2/7	2/6	2/6	2/5	2/5	2/5	1/4	1/4	1/4
0.65	3/8	3/7	2/6	2/5	2/5	2/5	2/5	1/4	1/4	1/4
0.66	3/7	2/6	2/6	2/5	2/5	2/5	2/5	1/4	1/4	1/4
0.67	3/7	2/6	2/6	2/5	2/5	2/5	1/4	1/4	1/4	1/4
0.68	3/7	2/6	2/6	2/5	2/5	2/5	1/4	1/4	1/4	1/4
0.69	3/7	2/6	2/6	2/5	2/5	2/5	1/4	1/4	1/4	1/4
0.70	3/7	2/6	2/6	2/5	2/5	2/5	1/4	1/4	1/4	1/4
0.71	3/7	3/6	2/5	2/5	2/5	2/4	2/4	2/4	2/4	2/4
0.72	3/7	3/6	2/5	2/5	2/5	2/4	2/4	2/4	2/4	2/4
0.73	3/7	3/6	2/5	2/5	2/5	2/4	2/4	2/4	2/4	2/4
0.74	3/7	3/6	2/5	2/5	2/5	2/4	2/4	2/4	2/4	2/4
0.75	3/7	3/6	2/5	2/5	2/4	2/4	2/4	2/4	2/4	2/4
0.76	4/7	3/6	2/5	2/5	2/4	2/4	2/4	2/4	2/4	2/4
0.77	3/6	3/6	2/5	2/5	2/4	2/4	2/4	2/4	2/4	1/3
0.78	3/6	3/6	3/5	3/5	2/4	2/4	2/4	2/4	2/4	1/3
0.79	3/6	3/5	3/5	3/5	2/4	2/4	2/4	2/4	2/4	1/3
0.80	3/6	3/5	3/5	3/5	2/4	2/4	2/4	2/4	2/4	2/3
0.81	3/6	3/5	3/5	2/4	2/4	2/4	2/4	2/4	2/3	2/3
0.82	4/6	3/5	3/5	2/4	2/4	2/4	2/4	2/4	2/3	2/3
0.83	4/6	3/5	3/5	2/4	2/4	2/4	2/4	2/4	2/3	2/3
0.84	4/6	3/5	3/5	2/4	2/4	2/4	2/4	2/4	2/3	2/3
0.85	4/6	3/5	3/5	2/4	2/4	2/4	2/4	2/3	2/3	2/3
0.86	4/6	3/5	3/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3
0.87	4/6	3/5	3/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3
0.88	4/6	3/5	3/4	3/4	3/4	3/4	3/4	2/3	2/3	2/3
0.89	4/6	4/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3	2/3
0.90	4/6	4/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3	2/3
0.91	5/6	4/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3	2/3
0.92	5/6	4/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3	2/3
0.93	4/5	4/5	3/4	3/4	3/4	3/4	2/3	2/3	2/3	2/3
0.94	4/5	4/5	3/4	3/4	3/4	2/3	2/3	2/3	2/3	2/3
0.95	4/5	4/5	3/4	3/4	3/4	3/3	3/3	3/3	3/3	3/3
0.96	4/5	4/5	4/4	4/4	4/4	3/3	3/3	3/3	3/3	3/3
0.97	5/5	5/5	4/4	4/4	4/4	3/3	3/3	3/3	3/3	3/3
0.98	5/5	4/4	4/4	4/4	4/4	3/3	3/3	3/3	3/3	3/3
0.99	5/5	4/4	4/4	4/4	4/4	3/3	3/3	3/3	3/3	3/3

B. Location Awareness

Knowing the location of any goods is very important at all times in warehousing management. Among all positioning technologies, the UWB positioning technology has a tendency to prosper in recent years due to its high precision, especially in industrial monitoring, safety emergency management, and plant staff management. Both TDOA and RSS can be used to obtain distance information (range measurements) between ANs and MT, and these two type of measurements are very easy to obtain. It is noted that the environment of warehouse is harsh and time varying, and the obstructions are randomly scattered. Therefore, the NLOS situation is unavoidable. NLOS identification and mitigation methods based on the statistical analysis of channels and waveforms [48], [49] are impractical because of the enormous measurement activities. The research reported in this article considers the situation in which NLOS measurements are unrecognizable. We propose an RRW algorithm, which can mitigate the NLOS error. Because the RRW algorithm mentioned in this article still uses the traditional Taylor algorithm [58], we will briefly review the Taylor algorithm before discussing it.

1) Taylor Algorithm: Let (x, y) be the real position of the MT. The coordinates (X_i, Y_i) are the known position of the ith AN, and $i = 1, 2, \ldots, Q$, Q indicates the number of ANs. According to the TDOA principle, the following equations are formed:

$$d_i - d_1 = m_{i1}c - e_i \quad (9)$$

where

$$d_i = \sqrt{(X_i - x)^2 + (Y_i - y)^2} \quad (10)$$

$$d_1 = \sqrt{(X_1 - x)^2 + (Y_1 - y)^2} \quad (11)$$

and

c: the speed of light;

e_i: the measurement error;

m_{i1}: the measured value, that is, the time difference between the arrival of the signal at the ith AN and first AN.

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Obviously, if all ANs receive signals at a certain moment, $M-1$ equations can be listed like (9). Expand the left-hand side of (9) in Taylor's series [58] and keep only terms below second-order

$$r_{i0} - r_{10} + a_{i1}\delta_x + a_{i2}\delta_y \approx m_{i1}c - e_i \tag{12}$$

where

$$r_{i0} = \sqrt{(X_i - x_0)^2 + (Y_i - y_0)^2} \tag{13}$$

$$r_{10} = \sqrt{(X_1 - x_0)^2 + (Y_1 - y_0)^2} \tag{14}$$

$$a_{i1} = \frac{X_1 - x_0}{r_{10}} - \frac{X_i - x_0}{r_{i0}} \tag{15}$$

$$a_{i2} = \frac{Y_1 - y_0}{r_{10}} - \frac{Y_i - y_0}{r_{i0}}. \tag{16}$$

(x_0, y_0) indicates the initial expansion point. Therefore, according to the least-squares method, the following conclusions can be obtained:

$$\boldsymbol{\delta} = \left(\boldsymbol{A}^T\boldsymbol{R}^{-1}\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\boldsymbol{R}^{-1}\boldsymbol{Z} \tag{17}$$

where

$$\boldsymbol{\delta} = \begin{bmatrix}\delta_x \\ \delta_y\end{bmatrix} \tag{18}$$

$$\boldsymbol{A} = \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \\ \vdots & \vdots \\ a_{Q1} & a_{Q2}\end{bmatrix} \tag{19}$$

$$\boldsymbol{Z} = \begin{bmatrix} b_{21} - (r_{20} - r_{10}) \\ b_{31} - (r_{30} - r_{10}) \\ \vdots \\ b_{Q1} - \left(r_{Q0} - r_{10}\right)\end{bmatrix} \tag{20}$$

$$b_{i1} = m_{i1}c. \tag{21}$$

Note that $\boldsymbol{R}$ is the covariance matrix of $\boldsymbol{e}$, and $\boldsymbol{e} = \begin{bmatrix}e_2 & e_3 & \dots & e_Q\end{bmatrix}^T$. Define $\mathbf{x} = [x, y]^T$ and $\mathbf{x}_0 = [x_0, y_0]^T$. Once the $\boldsymbol{\delta}$ is obtained, an iterative process is performed. Replace $\mathbf{x}_0 + \boldsymbol{\delta}$ with $\mathbf{x}_0$ and repeat the same process until the $\|\boldsymbol{\delta}\|$ value is less than a certain threshold δ_{th}. Here, $\|\cdot\|$ denotes the norm operation of a vector, and δ_{th} is a value close to 0 set in advance. Thence, the algorithm of TDOA is as follows.

2) RSS Residual Weighting Algorithm: A frequently reported model for the fading channel gives the mean dB received power at AN i that was transmitted from MT as

$$p_i = p_0 - 10n\log_{10}\left(\frac{d_i}{d_0}\right) - \zeta_i \tag{22}$$

where p_0 is the received power in dB at a reference distance d_0 and n is the path-loss exponent. The subscript i denotes the ith AN, $i = 1, 2, \dots, Q$; d_0 and d_i are the reference distance (here we set $d_0 = 1$) and the real distance from the ith AN, respectively. Thus, p_0 and p_i are the reference RSS corresponding to d_0 and the desired RSS (measured RSS). ζ_i represents the shadowing factor. Here, we assume that ζ_i is a normal variable with zero mean and variance σ_{dB}^2.

Algorithm 2 Taylor Algorithm

```
Init x_0 and δ_th.
δ ← 0
repeat
    x_0 ← x_0 + δ.
    Compute δ according to Equation (17).
until ‖δ‖ < δ_th
x ← x_0.
```

By the Taylor algorithm (see Algorithm 2), we can estimate the position $(\hat{x}, \hat{y})$ of the MT. Then the distance between MT and the ith AN can be estimated as

$$\hat{d}_i = \sqrt{\left(X_i - \hat{x}\right)^2 + \left(Y_i - \hat{y}\right)^2}. \tag{23}$$

Assuming that the propagation path between the MT and the ith AN is LOS, then the estimated RSS is

$$\hat{p}_i = p_0 - 10n\log_{10}\left(\frac{\hat{d}_i}{d_0}\right). \tag{24}$$

Then, we define $p_{\text{res},i}$ as the RSS residual for the ith AN

$$p_{\text{res},i} = \hat{p}_i - p_i. \tag{25}$$

Comparing (24) with (22), the $p_{\text{res},i}$ value is also estimated as the shadowing factor ζ_i

$$\zeta_i \approx p_{\text{res},i}. \tag{26}$$

Generally, we consider the absolute value of $p_{\text{res},i}$, so it is reasonable to regard $p_{\text{res},i}^2$ as the degree of obstruction of NLOS. Assuming that the number of ANs (i.e., Q) is greater than the minimum number (i.e., 3 for TDOA of 2-D) of ANs required for TDOA, we can get multiple estimated locations by different combination of ANs. We use $1 \sim Q$ to identify the index of Q ANs, and use the set symbol $\mathbf{S}$ to represent the set of all indices. In our RRW algorithm (see Algorithm 3), we will select a part of ANs to estimate the position in each round. After the estimation of each round, we will remove one of the ANs that is most likely NLOS from $\mathbf{S}$. Thus, to guarantee at least three ANs, we only need to calculate the $Q-2$ combinations, as the AN with the highest $p_{\text{res},i}^2$ value (the most likely NLOS) will be kicked out in each round except the first round. In the kth ($k = 1, 2, \dots, Q-2$) round, we calculate the estimated location $(\hat{x}_k, \hat{y}_k)$ of its ANs index combinations denoted as set $\mathbf{S}_k$. Obviously, the number of ANs in each round, i.e., the size of $\mathbf{S}_k$, is $Q - k + 1$. In addition, except for the last round ($k = Q - 2$), the $p_{\text{res},i}^2$ of each AN of its combination is calculated, and the AN with the highest $p_{\text{res},i}^2$ value will be kicked out in the next round. In order to evaluate the result $(\hat{x}_k, \hat{y}_k)$ of each round, we can use the sum of squares of RSS residuals to express the credibility of the result. The following equation gives the sum of squares of RSS residuals:

$$\text{RRES}_k = \sum_{i \in S_k} p_{\text{res},i}^2. \tag{27}$$

However, the size of $\mathbf{S}_k$ in each round is different. So, it is necessary to redefine a normalized RRES_k, $\widetilde{\text{RRES}}_k$, to alleviate

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Algorithm 3 RRW Algorithm

1: $\mathbf{S} \leftarrow \{1, 2, \ldots, Q\}$
2: **for** $k = 1$ to $Q - 2$ **do**
3: $\quad \mathbf{S}_k \leftarrow \mathbf{S}$.
4: $\quad$ Compute the $(\hat{x}_k, \hat{y}_k)$ by Taylor method (see Algorithm 2) using ANs whose index is in $\mathbf{S}_k$.
5: $\quad$ Compute $p^2_{res,i}$ for each AN, $i \in \mathbf{S}_k$.
6: $\quad$ Compute $\widetilde{RRES}_k$.
7: $\quad$ Find the AN index with the maximum value $p^2_{res,i}$ in $\mathbf{S}_k$, then kick it out from $\mathbf{S}$.
8: **end for**
9: Compute the final estimated location $\hat{\mathbf{x}}$ according to Equation (29).

the dependence on the size of $\mathbf{S}_k$

$$\widetilde{\mathrm{RRES}}_k = \frac{\mathrm{RRES}_k}{Size\ of\ \mathbf{S}_k} = \frac{\mathrm{RRES}_k}{Q - k + 1}. \tag{28}$$

The magnitude of the $\widetilde{\mathrm{RRES}}_k$ value can be used to characterize the quality of the result $(\hat{x}_k, \hat{y}_k)$. The smaller the $\widetilde{\mathrm{RRES}}_k$ value is, the lower the probability of NLOS in $\mathbf{S}_k$, and the more trustworthy this result is. So, we can weight each $(\hat{x}_k, \hat{y}_k)$ according to its $\widetilde{\mathrm{RRES}}_k$. In consequence, the final location estimate $(\hat{x}, \hat{y})$ is the linear combination of the $(\hat{x}_k, \hat{y}_k)$ weighted inversely to their $\widetilde{\mathrm{RRES}}_k$

$$\hat{\mathbf{x}} = \frac{\sum_{k=1}^{M-2} \hat{\mathbf{x}}_k \left(\widetilde{\mathrm{RRES}}_k\right)^{-1}}{\sum_{k=1}^{M-2} \left(\widetilde{\mathrm{RRES}}_k\right)^{-1}} \tag{29}$$

where $\hat{\mathbf{x}}$ and $\hat{\mathbf{x}}_k$ are both vectors

$$\hat{\mathbf{x}} = \left[\hat{x}, \hat{y}\right]^T \tag{30}$$

$$\hat{\mathbf{x}}_k = \left[\hat{x}_k, \hat{y}_k\right]^T. \tag{31}$$

In summary, assuming that the number of ANs is greater than 3 (i.e., $Q > 3$), the RRW algorithm is shown in Algorithm 3.

IV. Experiments

In this section, we will validate and evaluate two core algorithms through experiments. In addition, as an example, we also present a real working practice in the warehouse of Eletile company in Zhangjiagang, China.

A. Goods Sensing Experiments

The truth is, the reading rate p is a function of distance between antenna and RFID tag, and it is inversely proportional to distance. Fig. 5 is the reading rate of IV7 gotten from our experiment campaign. The points in Fig. 5 refer to Table IV. We also give the reading rate both on the side and back of the forklift in Table IV.

We installed two antennas up and down on the forklift, and the height difference between the two antennas is 70 cm. A single tag is suspended at varying distances in the same plane as the middle of the two antennas. For every 20-cm increment

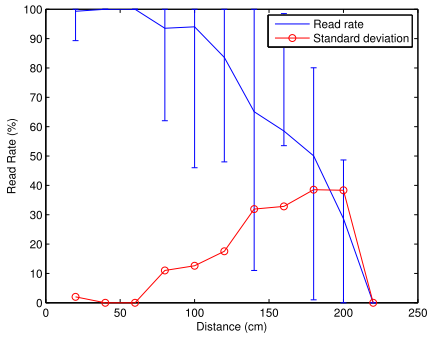


Fig. 5. Reading rate of IV7 at different distances.

TABLE IV
Reading Rate of IV7 at Different Distances

Distance(cm)	Reading rate(%)	Standard deviation σ(%)
20	99.30	2.00
40	100.00	0.00
60	100.00	0.00
80	93.50	11.00
100	94.00	12.60
120	83.57	17.52
140	65.08	31.90
160	58.52	32.84
180	50.66	38.50
200	28.63	38.33
220	0.00	0.00
50 [a]	0.30	18.00
50 [b]	0.00	0.00

[a] from the side of the forklift.
[b] from the back of the forklift.

of distance from the reader, we measured the reading rate of IV7. A reading rate is calculated every 100 interrogation cycles. Collect 50 reading rates at each distance, and then calculate the average reading rate, maximum and minimum reading rate, and standard deviation (i.e., σ) of the reading rate. Note also that, users can control the RF power level of IV7 for each antenna through basic reader interface (BRI) [59]. Here, we configure the RF level of each antenna to 27 dB according to the length of the forklift fork, which is about 120 cm in our case. Furthermore, it can be seen from Fig. 5 that the σ increases as the distance increases. It appears to be unstable when the distance exceeds 100 cm.

The filtering results of using respective M/N-K sliding window at different distances are shown in Fig. 6. When the goods leave the forklift completely, the K window always works, so here we mainly focus on the algorithm part of M/N. We can find that this method performs well within 120 cm [Fig. 6(a) and (b), the experimental results of less than 120 cm are similar] without a misjudgment (FN). However, due to the large standard deviation σ, misjudgment will occur more frequently when the distance is further increased [Fig. 6(c)]. There will be some FN results, under such circumstances. Fortunately, with this method, when FN occurs, the system will return to the correct state in a short time. In fact, if $\sigma = 0$, $p = 0.58$, and $\delta = 0.01$, compared with Fig. 6(c), the filtering performance

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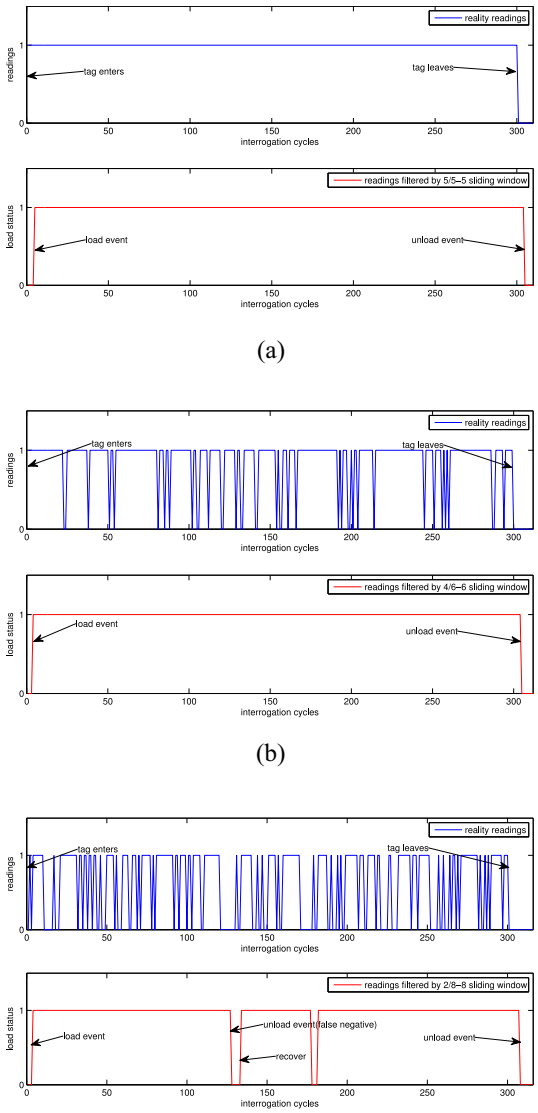


Fig. 6. Filtering results of IV7. (a) Distance = 20 cm, $\delta = 0.01$, $p = 0.99$, and $\sigma = 0.02$. (b) Distance = 120 cm, $\delta = 0.01$, $p = 0.83$, and $\sigma = 0.18$. (c) Distance = 160 cm, $\delta = 0.01$, $p = 0.58$, and $\sigma = 0.33$.

of M/N-K (referring to Table III, $M = 2$, $N = 8$, and $K = 8$) sliding window is not bad. We simulated this and gave the results in Fig. 7, which tells us that when $\sigma = 0$, the system can run almost error free in theory.

In practical applications, when we are familiar with the reading rate performance of RFID reader, we only need to use a fixed and reasonable M/N-K parameter. Since the reading rate of IV7 on the fork edge is above 70%, we can use $M = 3$, $N = 7$, and $K = 7$ (refer to Table III, $p = 0.70$, and $\delta = 0.01$) as the fixed sliding window parameter. The experimental filtering results of IV7 under fixed sliding window parameters are depicted in Fig. 8. Experiments show that our approach can handle the loading and unloading decision accurately using RFID readings.

In addition, we are interested in evaluating the loading event (i.e., the existence of the RFID tag) probability of the existence

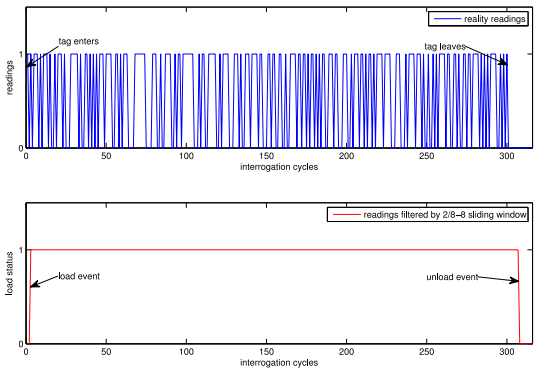


Fig. 7. Filter simulation, $\delta = 0.01$, $p = 0.58$, and $\sigma = 0$.

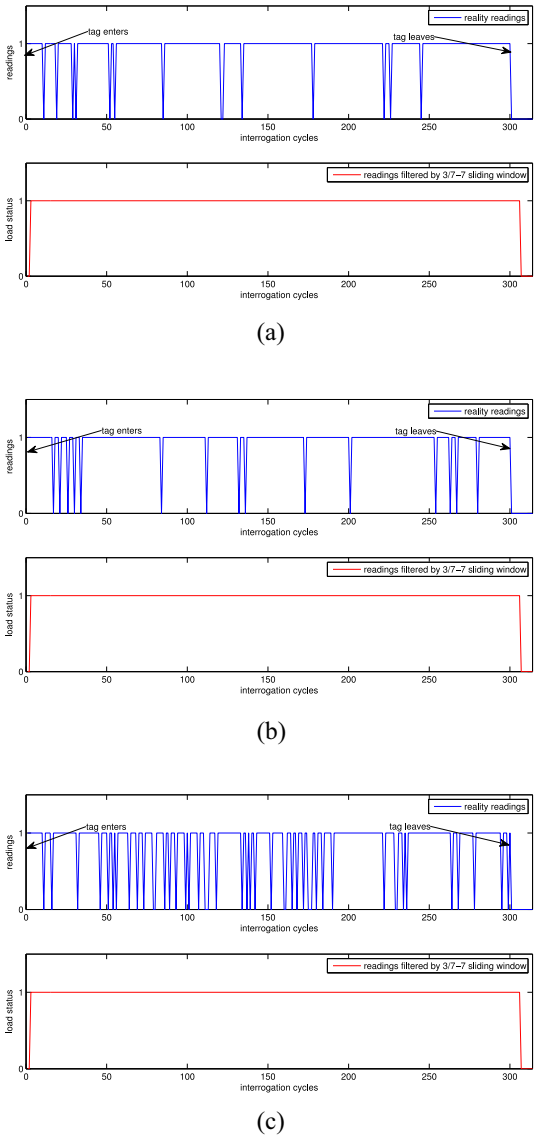


Fig. 8. Filtering results of IV7 using fixed M/N-K sliding window. (a) Distance = 80 cm, $p = 0.93$, and $\sigma = 0.11$. (b) Distance = 100 cm, $p = 0.94$, and $\sigma = 0.13$. (c) Distance = 120 cm, $p = 0.83$, and $\sigma = 0.18$.

of the RFID tag after a certain number of interrogation cycles. If we ignore or do not take "sliding of window" into consideration, this would be a very simple binomial distribution [60].

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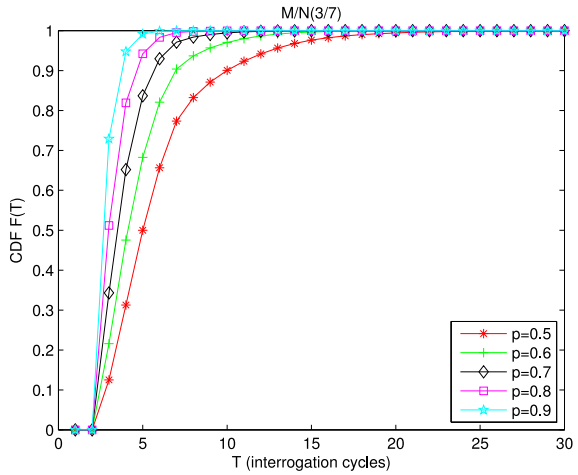


Fig. 9. $F(T)$ of $M/N(3/7)$.

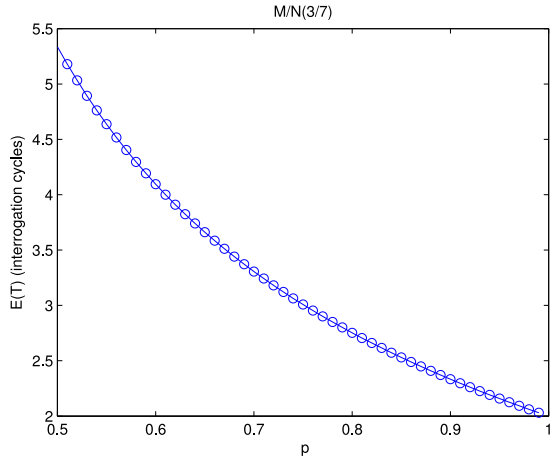


Fig. 10. $E(T)$ of $M/N(3/7)$.

Actually, the process of calculating acquisition probability is a Markov problem [57], [61], [62]. We use the symbol T to indicate the time (i.e., the number of interrogation cycles) at which the loading event occurs after the RFID tag appears on the forklift at time zero. Obviously, T is a random variable. The symbol $P(T)$ denotes the probability of a loading event occurring at time T. Thus, the cumulative distribution function (CDF) of T is $F(t) = P(T \leq t)$. With reference to the method of [57], we can easily obtain $F(T)$ (Fig. 9) and the expectation of T, $E(T)$ (Fig. 10), without considering σ. As can be seen from Fig. 9, above 90% of the loading event can be detected within ten interrogation cycles when $p \geq 0.5$. As can be seen from Fig. 10, the average (expectation of) detection cycles is <5.5 interrogation cycles when $p \geq 0.5$. As we all know, the average detection time = one interrogation cycle length × the average detection cycles. Since length of one interrogation cycle of IV7 is 25 ms (this can be easily obtained through software log), the average detection time is <137.5 (25 ms × 5.5) milliseconds when $p \geq 0.5$. These experimental results show that the proposed method has high availability and low latency.

Now we will analyze the *precision*, *recall*, *accuracy*, and *F-score* of the algorithm through simulation according to the measured parameters (Table IV). First, we give the following definitions.

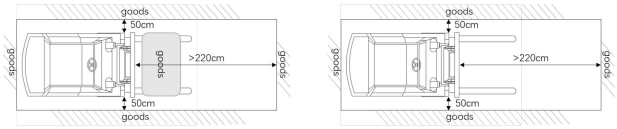


Fig. 11. Experimental scenario. (a) Forklift with goods. (b) Forklift without goods.

1) *True Positive (TP):* Forklift with goods correctly identified as forklift with goods.
2) *FP:* Forklift without goods incorrectly identified as forklift with goods.
3) *True Negative (TN):* Forklift without goods correctly identified as forklift without goods.
4) *FN:* Forklift with goods incorrectly identified as forklift without goods.

Refer to Fig. 11 for the experimental scenario. We assume that when the forklift is parked, the front of the forklift is always far away from (>220 cm) the goods on the ground (this is reasonable, we can strictly restrict the forklift driver in this regard). Therefore, we can only consider FP and TN caused by side goods and TP and FN caused by goods on forklift. We simulated a total of 10 000 times of experiments with goods on forklifts. The experimental results show that all 10 000 times can be correctly identified, that is, the number of times of TP is 10 000 (denoted as TP = 10 000), and the number of times of FN is 0 (denoted as FN = 0). Similarly, since loading and unloading events usually occur in pairs, we also simulated 10 000 times of experiments without goods on the forklift, but there were goods on the side of the forklift. The experimental results show that FP = 717 and TN = 9283. Therefore

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} = 93.3\% \tag{32}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} = 100\% \tag{33}$$

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}} = 96.4\% \tag{34}$$

$$F\text{-score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = 48.27\%. \tag{35}$$

In a nutshell, the M/N-K method can quickly, accurately, and efficiently determine the loading and unloading states of the goods on the forklift at low σ, avoiding the installation of additional equipment (e.g., photo sensor [13]) and hence saving costs.

B. RRW Experiments

In this section, we will evaluate the performance of the proposed RRW algorithm in a simulated environment. We first analyze the composition of the measurement error. Then, the path-loss exponent n is estimated by the RSS measured in the real warehouse environment. In addition, we list some other important parameters in the experiments. Finally, some performance comparisons among RRW, traditional Taylor

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TABLE V
POSITION OF ANS

AN index	coordinate [a]
1	(0,0)
2	(30,10)
3	(60,0)
4	(60,60)
5	(30,50)
6	(0,60)

[a] Unit:m.

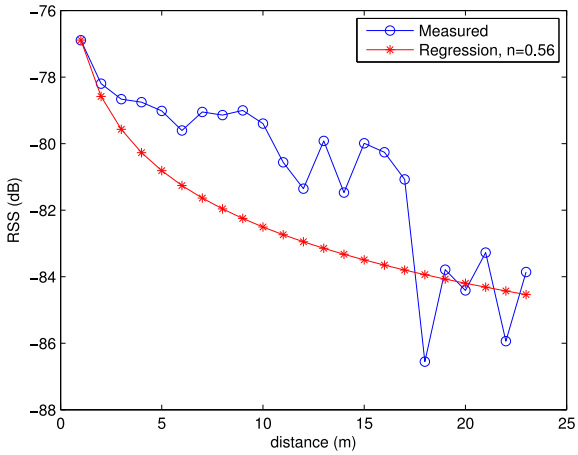


Fig. 12. RSS measurements and estimation of path-loss exponent n.

method [58], and Convex approximation method [53] (Convex for short) are given.

In our experiment, there are six ANs (see the red diamond mark in Fig. 13), which are deployed in a square area of 60 m × 60 m. Table V gives the coordinate of six ANs. We assume that the MT can communicate with all ANs anywhere in the area. This is consistent with the reality, that is, in general, UWB signals can travel up to about 100 m. For the NLOS environment, we assume that *NumNS* ANs are blocked each time. But as to which ones are blocked, it is totally random. Considering that the measurement results are easily corrupted by NLOS, the measurement error is modeled as follows:

$$e_i = e_i' + e_i'' \tag{36}$$

where

$$e_i' \sim \mathcal{N}\left(0, \sigma_d^2\right) \tag{37}$$

$$e_i'' = \begin{cases} 0, & \text{if LOS} \\ \epsilon_i \sim \text{Exp}(\lambda), & \text{if NLOS.} \end{cases} \tag{38}$$

As (36) shows, the measurement error e_i in (9) contains two types of errors: 1) measuring error e_i' and 2) NLOS error e_i''. The measuring error is normally distributed with a mean of 0 and a variance of σ_d^2. Here, we assume $\sigma_d = 0.1$ m. If there is no blocking between the ith AN and the MT, then there is no e_i'', otherwise NLOS error is exponentially distributed with mean λ.

In order to obtain the path-loss exponent n, we measured the RSS value of the MT every 1 m. Furthermore, according to (22), and regression analysis by the least-squares method, the final estimated n is 0.56. RSS measurements and regression results are shown in Fig. 12. Meanwhile, we define 1 m as the reference distance d_0, and the RSS p_0 measured here is -76.89 dB. All the parameters used in the simulation experiments are listed in Table VI.

TABLE VI
SIMULATION PARAMETERS

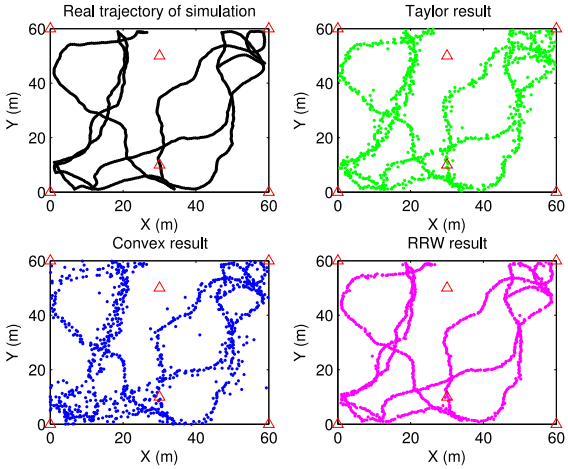
Parameter	Value	Explain
Q	6	the number of ANs
σ_d	0.1 m	$e_i' \sim \mathcal{N}(0, \sigma_d^2)$
λ	1m-20m	$e_i'' \sim Exp(\lambda)$, if NLOS; 0, if LOS.
$NumNS$	1-6	number of NLOS
n	0.56	path loss exponent
p_0	-76.89 dB	the received power at d_0
d_0	1 m	the reference distance

As depicted in Fig. 13, we simulate the true trajectory (1000 points) of a moving object, and give the trajectory results calculated by the traditional Taylor algorithm, Convex algorithm, and RRW algorithm after measuring error and NLOS error are added to measurements. It can be seen from Fig. 13 that the RRW algorithm is much better than the Taylor algorithm and Convex algorithm as its result is closer to the true trajectory. While the trajectories restored by the Taylor algorithm are ambiguous. The Convex method is a method that can mitigate NLOS error, but it does not perform as well as RRW. In addition, as the value of λ increases, the number of points representing the RRW result that are obviously outside the trajectory increases. This is also the reason for the increase in RMSE, which we will talk about later. Note that these results (Fig. 13) are based on the parameters in Table VI, where the value of *NumNS* is 2.

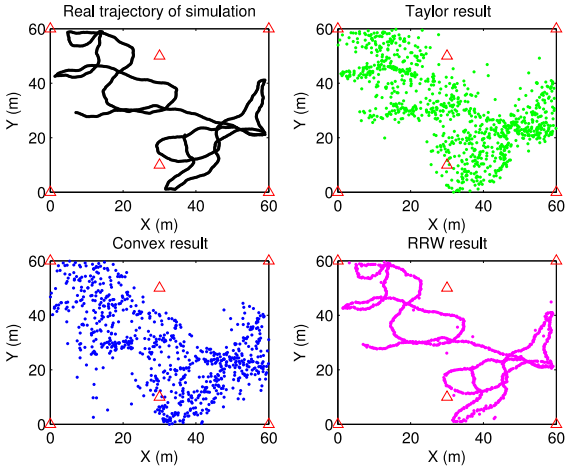
Fig. 14 shows the comparisons of empirical CDF of the Taylor error, Convex error, and RRW error under different λ. When λ becomes larger [see Fig. 14(c) and (d)], compared with Taylor, Convex does eliminate some NLOS errors, but the effect is not as good as RRW. Thus, these results imply that the accuracy of RRW is significantly higher than that of Taylor and Convex. From Fig. 14, we also find that the increase of λ has little effect on the CDF of the RRW error, which shows that the RRW algorithm is stable and reliable.

Next, we examine the impact of *NumNS* on the performance of the RRW algorithm. Fig. 15 gives the comparisons of empirical CDF of the Taylor error, Convex error, and RRW error under different *NumNS*. Fig. 15 indicates that when $NumNS \leq Q - 3$, the accuracy of the RRW algorithm is obviously higher than the Taylor algorithm and Convex algorithm. But when there are fewer LOSs, that is, just less than the minimum number of ANs required (for 2-D positioning, this also means that when *NumNS* is greater than $Q - 3$), the performance of the RRW algorithm is not much better than Taylor and Convex.

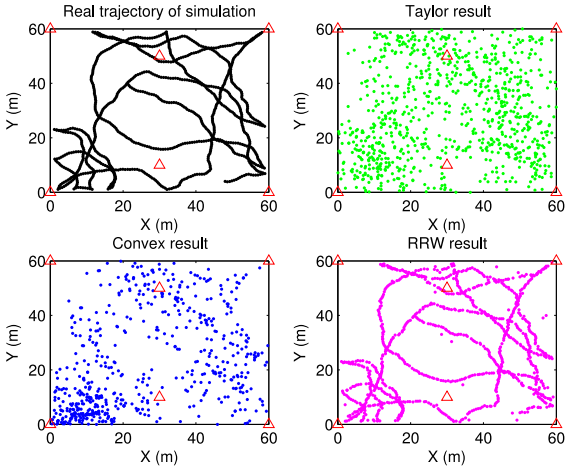
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(a)

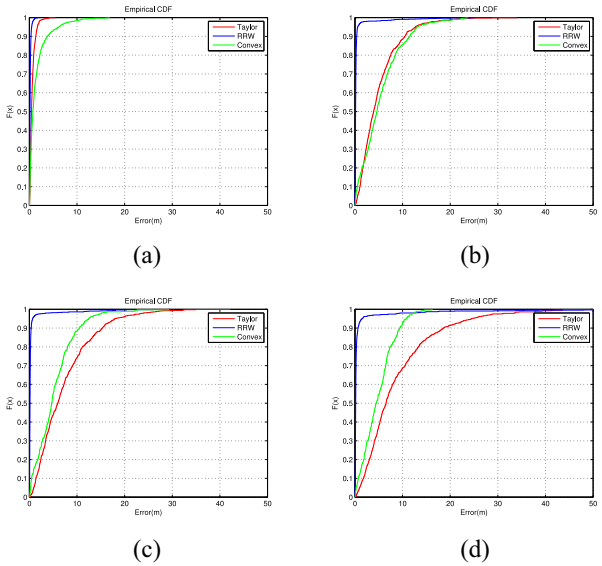


(b)



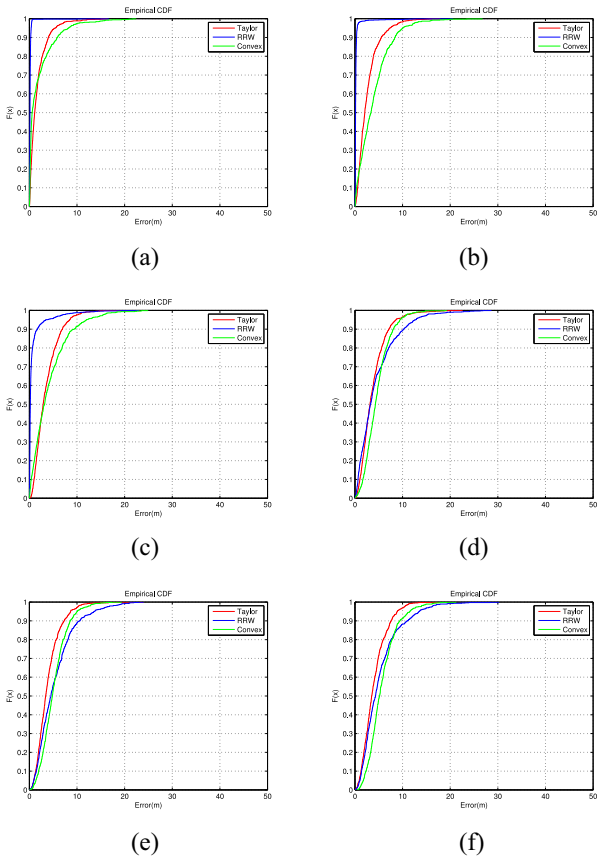
(c)

Fig. 13. Positioning results of Taylor, Convex, and RRW. *NumNS* = 2. (a) λ = 1. (b) λ = 5. (c) λ = 10.



(a) (b) (c) (d)

Fig. 14. CDF of the Taylor error, Convex error, and RRW error under different λ. *NumNS* = 2. (a) λ = 1. (b) λ = 5. (c) λ = 10. (d) λ = 20.

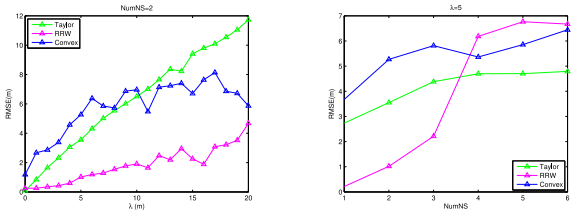


(a) (b) (c) (d) (e) (f)

Fig. 15. CDF of the Taylor error, Convex error, and RRW error under different *NumNS*. λ = 5. (a) *NumNS* = 1. (b) *NumNS* = 2. (c) *NumNS* = 3. (d) *NumNS* = 4. (e) *NumNS* = 5. (f) *NumNS* = 6.

Finally, comparisons of the RMSE of these three algorithms are depicted in Fig. 16. Fig. 16(a) is the RMSE results under different λ, and Fig. 16(b) is the RMES results under different *NumNS*. It is noted [Fig. 16(a)] that when λ becomes larger, Convex shows a stronger NLOS mitigation capability than Taylor, but still not as good as RRW. As can be seen in Fig. 16(b), when *NumNS* becomes larger (*NumNS* > $Q - 3$), the accuracy of RRW decreases significantly. However, on the

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(a) (b)

Fig. 16. RMSE. (a) Under different λ. (b) Under different *NumNS*.



Fig. 17. Warehouse of Eletile company in Zhangjiagang, China.

whole, the accuracy of RRW is significantly higher than that of Taylor and Convex. It can be seen from Fig. 12 that the closest to the real trajectory is the result of RRW. The accuracy of the RRW algorithm decreases with the increase of NLOS. It can be seen from Fig. 12 that the main reason for the decline of RMSE is that some points are obviously out of actual trajectory. Therefore, we can further study the methods to eliminate these outliers in the future.

C. Example

In this section, we introduce the system that has actually been implemented in the warehouse in Eletile Company of Zhangjiagang, China. This is a company that makes plastic flooring. The system was deployed due to the need to improve the level of warehousing management. The warehouse in this project is about 5000 m^2, and a total of 22 ANs deployed. The height of ANs is 4 m, and the height of the goods is between 2 and 3 m. Therefore, UWB signal may be blocked when MT approaches the goods. We also have reformed four forklifts, and the velocity of forklift is no more than 15 km/h. The ERP system of the company made use of the information provided by this intelligent system to conduct goods inquiries, goods inventory, and goods dispatch.

Fig. 17 is the picture of the warehouse. One of the ANs is marked in the figure. There is also a large working screen in the picture. The current situation of the warehouse is on the screen via the positioning system and ERP system. Fig. 18(a) shows the UHF passive RFID tags used in the project. As shown in Fig. 18, the RFID tags are attached to the surface of the goods package.

Fig. 19 is a screenshot of the location of the goods in the warehouse at a certain time. The long strips in the picture



(a) (b)

Fig. 18. RFID tags and goods used in the warehouse. (a) RFID tags. (b) Goods with RFID tags.

TABLE VII
POSITION OF ANS IN WAREHOUSE OF ELETILE COMPANY

AN Name	coordinate [a]
CC19	(121.55,7.99)
CC20	(137.55,7.99)
CC23	(129.20,20.04)
AA24	(154.70,20.04)
CC26	(121.55,35.09)
CC27	(137.85,35.09)
CC21	(163.05,7.99)
CC28	(163.35,35.09)
CC25	(180.2,20.04)
CC22	(187.35,7.99)
CC29	(187.35,35.09)
CC30	(189.51,7.88)
CC37	(192.37,34.83)
CC34	(199.38,18.95)
CC38	(209.57,33.00)
CC31	(206.72,6.06)
AA35	(224.74,16.26)
CC32	(232.07,3.37)
CC39	(234.93,30.32)
CC36	(250.10,13.58)
CC33	(256.94,0.73)
CC40	(256.94,0.73)

[a] Unit:m.

are the parking lots of goods in the warehouse. The red dots indicate the location of the ANs (all 22 ANs are listed in Table VII), and the yellow dots indicate where the goods are currently parked. From the figure, we can see that most of the goods are parked in the lots, and it is consistent with the fact. Some of the goods in the figure are parked outside the parking lots, which partially is caused by positional deviations, but mainly by the behavior of the forklift driver temporarily moving the goods.

Fig. 20 illustrates the presentation of results of the system. We can see that the map has records of the loading and unloading locations of the goods, and also the track of the forklift. There is also a video screenshot in the figure. This

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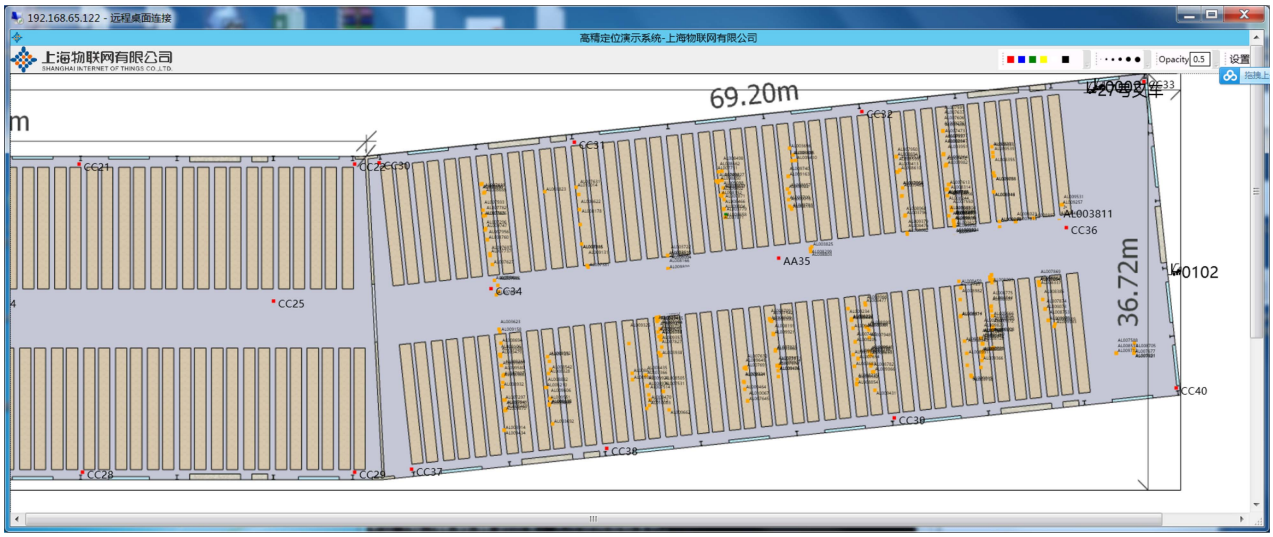


Fig. 19. Screenshot of location of goods in Warehouse of Eletile company in Zhangjiagang, China.

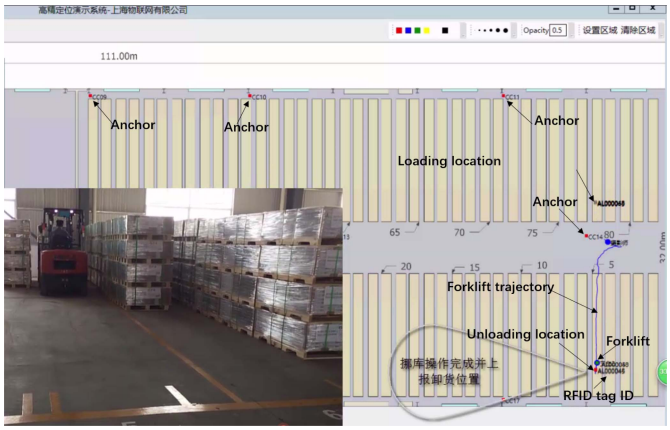


Fig. 20. Presentation of results.

really reflects the position and working state of the forklift. The experimental results demonstrate that in such a warehouse environment the system can achieve positioning accuracy less than 0.5 m.

In short, this project demonstrates the feasibility of the system.

V. Conclusion

In this article, we proposed an intelligent warehousing management system, including RFID and UWB technologies, and it makes a tradeoff between cost and accuracy of positioning. The designed system includes two core sensing functions: 1) goods sensing and 2) location awareness. Aiming at goods sensing, a novel algorithm named M/N-K sliding window was proposed to identify goods loading and unloading on a forklift. Experiments demonstrated that the M/N-K sliding window algorithm can quickly, accurately, and efficiently identify the loading and unloading status of goods. The method is simple and easy to implement, and hence it is very suitable for the embedded devices. For location awareness, we proposed a novel algorithm based on the RRW. Simulation experiments demonstrated that the novel algorithm can mitigate the NLOS error substantially compared with the Taylor algorithm and Convex algorithm. Thus, the RRW algorithm is suitable for such a harsh environment as warehouse. We deployed the prototype in a warehouse of a company. In summary, the system design is feasible. In our future work, we plan to gain a deeper insight into the algorithm and performance of the system in different types of warehouse, and build a more intelligent and widely adaptable intelligent warehousing system.

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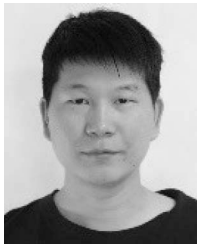


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